TEJAL GOYAL

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SUMMARY

Applied Data Science MEng and Software Engineer Co-op building backend services, AI agent tooling, and test automation in Python, TypeScript, C#, and SQL on Azure. Shipped an Anthropic API service that auto-resolved 50 of 83 records and eliminated 3+ hours of weekly manual matching.

EDUCATION

MEng Applied Data Science *Jan 2025 – Dec 2026*
University of Victoria, BC *GPA: 8.82/9.0*
Relevant Coursework: Algorithms and Data Models; Systems for Massive Datasets; Artificial Intelligence; Data Analysis and Pattern Recognition

BEng Electronics and Computer Engineering *2020 – 2024*
Thapar Institute of Engineering & Technology, Patiala, India *GPA: 8.9/10.0*
Relevant Coursework: Data Structures and Algorithms; Object-Oriented Programming; Database Management Systems; Operating Systems

WORK EXPERIENCE

Software Engineer Co-op *May 2026 – Dec 2026*
British Columbia Investment Management Corporation (BCI), Victoria, BC

- Shipped a vendor onboarding service on the Anthropic API that resolves counterparty codes against an existing hierarchy with a confidence score and rationale per record, eliminating 3+ hours of weekly manual matching and auto-resolving 50 of 83 records
- Build an AI-assisted Playwright regression suite for an Appian application, collapsing 40+ logged hours of manual click-through testing per cycle into a few clicks, with live-watch runs and a report of screenshots and recordings
- Traced a report-to-source data discrepancy through a migrated stack of cloud storage, a semantic model, and layered transformations, isolating records dropped for a missing identifier and writing a two-key matching function that recovered them
- Prototyped Python MCP (Model Context Protocol) servers exposing SQL Server and Azure SQL Managed Instance to AI agents, proving out on-demand agentic querying of enterprise financial data and informing the team's platform direction
- Contribute to a flagship financial system rebuilt from a vendor-locked .NET/WPF stack into a React application on Azure with GitHub Copilot, removing licensing risk and enabling cloud-native deployment
- Replaced a manual quarterly routine of running database queries, verifying feed completeness, and chasing support on gaps with a Power Automate flow that validates 12 feed-months and branches to the finished report or an itemized alert
- Coordinate user stories with product owners in a Scrum/Kanban hybrid using Jira and Confluence

Cybersecurity and Risk Co-op *Jan 2026 – Apr 2026*
British Columbia Investment Management Corporation (BCI), Victoria, BC

- Developed a PowerShell tool authenticating against a REST API, walking a page tree, and parsing HTML into structured exports, cutting a manual data extraction from 3 hours to 10 minutes

Research Intern, Federated Learning *Jan 2024 – Jul 2024*
Indian Institute of Technology Ropar, Punjab, India

- Engineered a modular federated learning framework in PyTorch with configurable client fraction, batch size, and local epochs, training ResNet and VGG across 4 datasets at scales up to 1M simulated clients
- Built a utility to programmatically convert TensorFlow code to PyTorch, cutting cross-framework integration from 2 days of manual rewriting to hours

PROJECTS

Dime: AI Expense Tracker *May 2026*
Next.js 15, React 19, TypeScript, Cloudflare Workers, Supabase, Google Gemini | dime.tgoyal.me

- Built and deployed an AI expense tracker on Next.js 15, React 19, and Supabase, integrating Google Gemini for natural-language expense parsing and real-time categorization
- Implemented auth with Supabase SSR cookie management, per-route verification on every API endpoint, and Row Level Security enforcing isolation at the PostgreSQL level

Claubi: AI Coding Agent *Apr 2026*
Rust, Tokio, Ollama, Serde, async-trait | github.com/tejalgoyal2/claubi

- Architected a local-first AI coding agent in Rust with a Planner-Executor-Reviewer pipeline, routing tasks to local models via Ollama with NDJSON streaming over async mpsc channels
- Implemented a deny-first permission model inspired by AWS IAM, routing file writes through a permission engine and validating shell commands against an allowlist, with append-only JSONL audit logging

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, TypeScript/JavaScript, C#, Rust, SQL, PowerShell, Bash, HTML/CSS
Backend & Data: .NET 8, Node.js, REST APIs, PostgreSQL, SQL Server, MongoDB, Redis, Supabase, Databricks, Azure Data Factory
AI & Agents: Anthropic API, Claude Code, MCP (Model Context Protocol), LLM Agents, RAG, Google Gemini API, PyTorch, Ollama
Frontend, Cloud & DevOps: React, Next.js, Azure, AWS, Cloudflare Workers, Docker, Kubernetes, Git, CI/CD, Playwright, Linux